

Transformers Cross-Attention Audio-Visual Fusion

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Abstract

Multimodal learning has become increasingly important within modern artificial intelligence systems, particularly in large multimodal models that jointly process visual, textual, and audio information. Recent multimodal architectures increasingly employ pre-trained modality-specific encoders to extract contextual latent representations prior to fusion, enabling richer preservation of acoustic and temporal information within speech signals. This work proposes a transformer-based audio-visual fusion pipeline for multimodal object recognition using the SpokenCOCO dataset. Pre-trained Vision Transformer and Whisper encoders were employed to extract modality-specific visual and audio embeddings, which were subsequently fused using a bidirectional cross-attention mechanism. The proposed architecture utilised frozen pre-trained encoders alongside lightweight downstream classifiers to evaluate the effectiveness of multimodal contextual fusion. Experimental evaluation demonstrated that multimodal fusion consistently outperformed single-modality representations across all evaluation metrics. These findings demonstrate the effectiveness of direct audio-visual fusion for multimodal contextual understanding and provide a foundation for future transformer-based multimodal grounding and object recognition systems.

Keywords: computer vision, machine learning, object classification, audiovisual and multimedia systems

1 Introduction

Transformer-based architectures have become the foundation of modern artificial intelligence (AI) systems, such as Large Language Models (LLMs) and Large Multimodal Models (LMMs). Early transformer-based language models such as BERT [Devlin et al., 2019] and GPT3 [Brown et al., 2020] demonstrated that transformer architectures can model long-range contextual relationships through self-attention mechanisms [Vaswani et al., 2017]. More recently, transformers have been extended beyond single-modality language processing towards multimodal learning systems capable of jointly processing text, images, audio and speech [Wu et al., 2024]. Several recent multimodal transformer architectures have demonstrated the effectiveness of combining multiple modalities for contextual understanding and generative tasks. Flamingo [Alayrac et al., 2022] introduced a visual-language model capable of conditioning Chinchilla language models [Hoffmann et al., 2022] on visual inputs, while ViCA [Liu et al., 2026] employed both self-attention and cross-attention mechanisms for visual-text reasoning. Beyond visual-language learning, transformer-based multimodal systems have also been applied to audio-visual tasks. Llama-AVSR [Cappellazzo et al., 2025] proposed an audio-visual speech recognition framework using LMMs while Macaw-LLM [Lyu et al., 2023] demonstrated text generation using images, audio and text inputs. These architectures highlight the increasing importance of multimodal learning within modern AI systems. Multimodal learning aims to model interactions between different data modalities in a manner similar to how humans integrate sensory information to develop contextual awareness of their environment [Shu et al., 2025]. By jointly processing multiple modalities, AI systems can exploit complementary information between data sources, allowing one modality to compensate when another is noisy, ambiguous, or incomplete. In audio-visual learning, combining speech and visual information can improve contextual understanding and robustness within recognition tasks [Ye et al., 2024]. To combine information from multiple modalities, multimodal systems employ fusion methods at different stages of the learning pipeline. Feature fusion combines modality representation after feature extraction but prior to classification [Li and Tang, 2026] while output fusion combines

predictions or outputs after classification [Li and Tang, 2026]. Common multimodal fusion methods include concatenation, [Gao et al., 2020] linear, [Li et al., 2017, Wu and Han, 2018] non-linear [Charte et al., 2018] and attention-based fusion approaches, such as cross-attention [Qian et al., 2023]. Among these approaches, transformer-based cross-attention mechanisms have become increasingly popular due to their ability to model contextual relationships between modalities. Recent multimodal architectures increasingly employ pre-trained modality-specific encoders to extract contextual latent representations prior to fusion [Wu et al., 2024]. Within audio-visual learning, transformer-based speech encoders provide temporally contextualised audio embeddings that preserve acoustic and prosodic information beyond textual transcriptions alone [Shu et al., 2025]. Similarly, transformer-based visual encoders produce contextual patch-level representation capable of modelling spatial relationships within images. These representations provide rich modality-specific contextual information that can be subsequently fused for downstream multimodal learning tasks.

This work investigates transformer-based audio-visual fusion using speech and visual embeddings extracted from Whisper and Vision Transformer pre-trained encoders. The proposed approach employs bidirectional cross-attention mechanisms to model contextual interactions between audio and visual modalities prior to fusion. Early feature-level fusion is performed by combining contextualised modality representations after bidirectional cross-attention, enabling the model to jointly learn relationships between speech and visual information. This paper contributes: a transformer-based multimodal fusion pipeline combining pre-trained ViT and Whisper encoders using cross-attention fusion and an evaluation of frozen pre-trained encoders with lightweight downstream classifiers.

2 Datasets

Many widely used computer vision datasets are single-modality or text-image based rather than speech-image based. Examples include the Large Vocabulary Instance Segmentation (LVIS) dataset [Gupta et al., 2019], which contains images for large-scale object recognition, and the GQA dataset [Hudson and Manning, 2019], which provides image-question-answer pairs for visual reasoning tasks. To enable speech modality learning, the dataset would either contain recorded speech, which can be expensive and time-consuming to create, or use Text-to-Speech (TTS) models to generate synthetic audio using the text captions from pre-existing text-image datasets such as SpeechCOCO [Havard et al., 2017] or how [Yang et al., 2025] created synthesised speech data from GQA textual questions and answers, using TTS model SpeechT5 [Ao et al., 2022]. For this work, datasets containing real human speech were preferred over synthetically generated speech datasets. Although TTS-generated datasets provide a scalable method for introducing the speech modality, synthetic speech often lacks speaker diversity or natural variation, unlike human speech [Hsu et al., 2021]. Based on these selection criteria, the following datasets were selected for use in this research.

MSCOCO [Lin et al., 2014] is an image-caption dataset containing 123,287 images and 605,495 captions, where each image is paired with 5 human-written captions. The dataset contains annotations for 80 object categories, with multiple object instances potentially present within a single image, as seen in Figure 1. The dataset additionally provides bounding box annotations for object instances present within the images.

SpokenCOCO [Hsu et al., 2021] is an additional dataset that contains spoken audio recordings corresponding to the textual captions from MSCOCO. These speech samples were collected through Amazon Mechanical Turk by displaying the caption text for the speakers to read out loud. The dataset contains recordings from 2353 speakers, primarily consisting of American English speakers, and has approximately 742 hours of spoken audio. The spoken captions are aligned with their corresponding MSCOCO images and textual captions through shared image identifiers provided within the dataset metadata. A challenge within both datasets is the severe class imbalance across object categories. For example, the *hair dryer* category contains only 198 images, whereas the *person* category contains 66,808 images. To enable computationally efficient experimentation while reducing the effects of severe class imbalance, a balanced subset of SpokenCOCO was constructed. The balancing process involved randomly selecting image-audio pairs containing at least one instance of a target class until each category contained 1000 training samples, 500 validation samples, and 500 testing samples. As MSCOCO images may contain multiple object categories simultaneously, selected samples were not restricted to

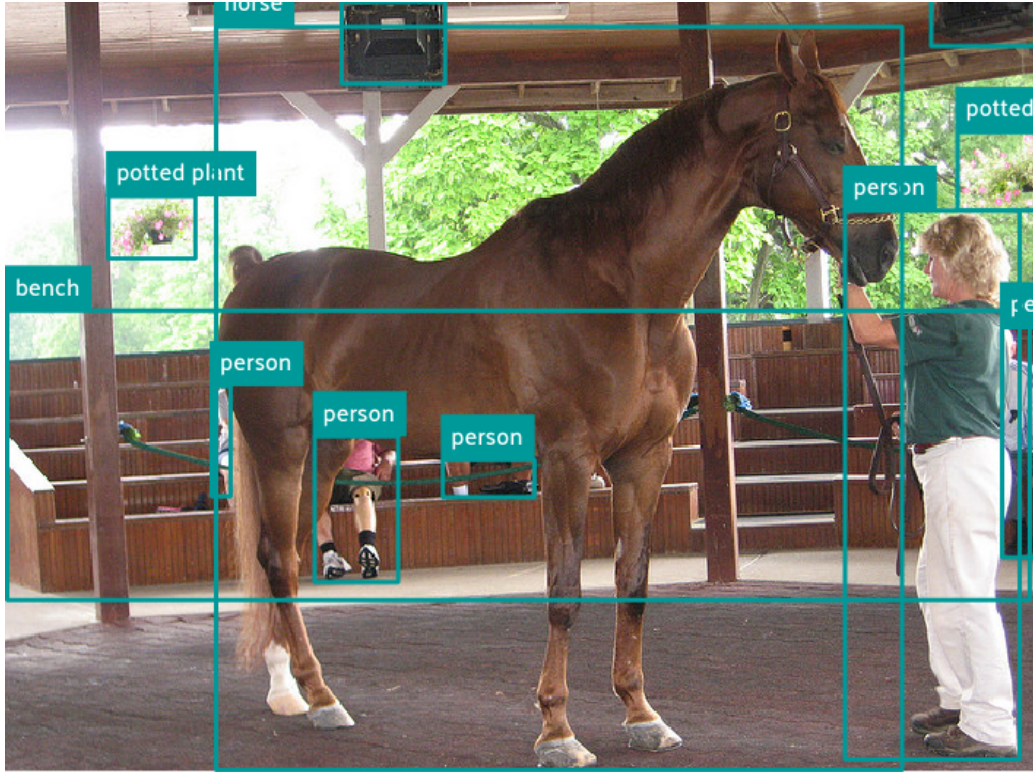


Figure 1: Example image from the MSCOCO dataset with its associated caption: "A woman interacts with a horse in a stable."

single-label scenes. 8 object classes were randomly selected from categories containing at least 2000 available samples within the original dataset. The selected object classes were *person*, *backpack*, *airplane*, *skis*, *kite*, *baseball glove*, *TV*, and *clock*. Classes with insufficient representation were excluded to ensure consistent sampling across training, validation, and testing splits. This process resulted in a balanced subset containing 16,000 paired image-audio pair samples across training, validation, and testing splits. The balanced subset provides a controlled experimental setting for evaluating transformer-based audio-visual fusion while reducing the influence of severe class imbalance present in the original datasets. The key statistics of the original COCO datasets and the balanced subset are summarised in Table 1.

Dataset	Images	Text Captions	Audio Captions	Classes	Speakers	Class Balance
MSCOCO	123,287	605,495	–	80	–	Imbalanced
SpokenCOCO	123,287	605,495	592,187	80	2,353	Imbalanced
Balanced Subset (Ours)	16,000	16,000	16,000	8	–	Balanced

Table 1: Summary of MS-COCO, Spoken-COCO and balanced subset datasets.

3 Methodology

The methodology in this work will cover multimodal audio-visual learning and classification. Several key stages include modality-specific feature extraction, bidirectional cross-attention between extracted audio and visual representations, contextualised feature fusion, and downstream classification. An overview of the proposed methodology is shown in Figure 2.

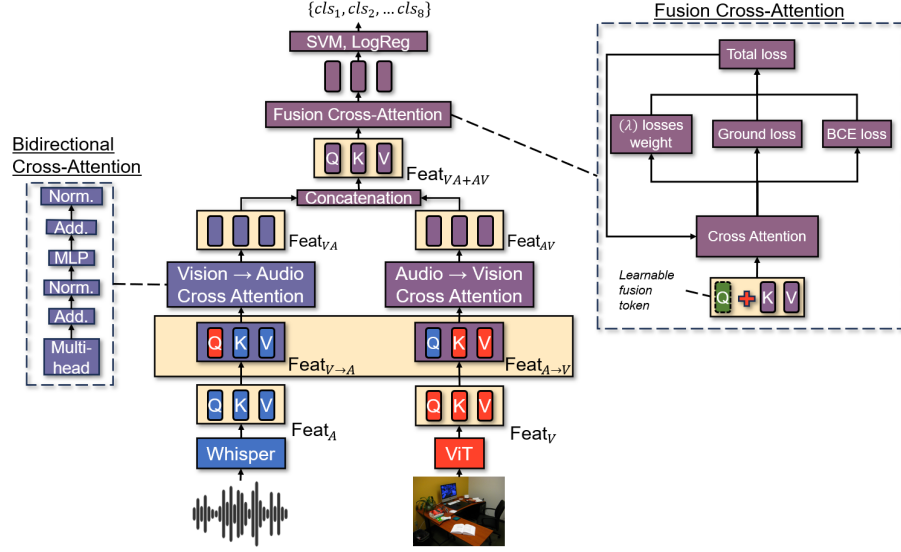


Figure 2: Overview of the proposed audio-visual fusion pipeline. Speech and image inputs are encoded using Whisper and ViT encoders to obtain modality-specific features (audio=A, visual=V). Bidirectional cross-attention modules perform audio-to-vision and vision-to-audio contextualisation ($\rightarrow$). The contextualised features are concatenated (+) and processed by a second fusion cross-attention module to produce a joint multimodal embedding for object classification. During training, classification and grounding losses are jointly optimised.

Audio-Visual Feature Extraction To obtain the representations used for multimodal fusion, pre-trained transformer-based encoders were employed for both the visual and audio modalities. Separate modality-specific encoders were used to preserve modality-specific feature representations prior to fusion.

Vision Transformer (ViT) [Dosovitskiy et al., 2021] was employed as a visual modality encoder. The model was pre-trained on ImageNet-22k [Ridnik et al., 2021]. Input images were processed and resized using Hugging Face `AutoImageProcessor` to match the ViT image size requirement of 224×224 resolution. After preprocessing, images were partitioned into non-overlapping 16×16 patches to produce $14 \times 14 = 196$ patch tokens with an additional learnable classification token appended, resulting in a total sequence length of 197 tokens. The image patches produced are flattened and linearly projected into an embedding vector before the positional embeddings are added to preserve spatial information. Embedded patch tokens are processed through the transformer encoder using self-attention mechanisms.

Whisper [Radford et al., 2023] was utilised as the audio modality encoder. Whisper is a transformer-based speech model pre-trained on its original dataset of 680k hours of labelled multilingual speech data. The "small" model variant was selected based on hardware constraints compared to other models and is a compromise of computational size and representational performance. Audio waveforms were first converted to floating-point tensors and resampled to 16 kHz, matching the input sampling rate expected by Whisper. The Hugging Face `WhisperProcessor` was then used to convert the raw audio waveforms into log-Mel spectrogram input features. During batch processing, padding was applied to produce fixed-size inputs, while longer audio sequences were truncated. Processed input features were passed through the Whisper encoder, and the final hidden states were extracted as the audio feature representation. Similar to ViT, the Whisper encoder applies self-attention to model contextual relationships across temporal audio embeddings. However, rather than modelling spatial relationships between image patches, Whisper captures contextual dependencies between speech representations across the audio sequence.

Both transformer encoders utilise the self-attention mechanism defined as follows [Vaswani et al., 2017]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where Q , K , and V represent the query, key, and value matrices, respectively, and d_k denotes the dimensionality

of the key vectors used for scaling. The Q matrix represents the contextual information each token seeks, the K matrix represents the semantic information contained within each token, and the V matrix contains the contextual representations that will be aggregated after attention weighting. The attention operation computes similarity scores between tokens, enabling the model to selectively aggregate contextual information across the input sequence. Following feature extraction, the pipeline produces two sequences of contextual embeddings: visual patch embeddings from ViT and temporal audio embeddings from Whisper. These embeddings are feature matrices, X_a and X_v for audio and visual matrices, respectively, as seen in Figure 2.

Both encoders were frozen to prevent updates to the pre-trained weights while reducing the training complexity of the overall pipeline and retaining the strong generalised representations learned from pretraining these models [Cappellazzo et al., 2025]. Each encoder produces embeddings with a feature dimensionality of 768, allowing compatibility during the cross-modal fusion process.

Bidirectional Cross-Attention To enable interaction between the extracted X_a and X_v matrices, bidirectional cross-attention fusion is used [Gao et al., 2024]. The purpose of this fusion stage is to incorporate contextual information from one modality matrix into the other, creating feature representations which are jointly informed of both speech and visual content. Unlike self-attention seen in the feature extraction using the transformer-based encoders, cross-attention allows tokens from one modality to attend to the tokens in the other modality. For this pipeline, two cross-attention mechanisms are used bidirectionally: Audio-to-Visual (A2V) – audio features attend to visual features; Visual-to-Audio (V2A) – visual features attend to audio features.

Cross-attention enables both modalities to become mutually context-aware. The audio representation emphasises visually relevant information, while the visual representations can incorporate contextual cues derived from speech. For further explanation, take the A2V cross-attention operation, where X_a is projected into query representations while X_v is projected into the key and value representations. The audio embeddings determine which visual tokens should receive attention. The resulting representation produces audio-informed visual contextualisation by allowing audio embeddings to selectively attend to semantically relevant visual regions. The attention operation for the i -th attention head is defined as [Vaswani et al., 2017]:

$$\text{head}_i = \text{softmax} \left(\frac{(X_a W_i^Q)(X_v W_i^K)^T}{\sqrt{d_k}} \right) (X_v W_i^V) \quad (2)$$

where W_i^Q , W_i^K , W_i^V denote the learnable projections for the query, key, and value representations of the i -th attention head, respectively, and d_k represents the dimensionality of the key vectors for scaling. The softmax operation normalises the similarity scores between audio queries and visual keys.

The cross-attention mechanisms employ multi-head attention with 8 attention heads, consistent with the original attention mechanism [Vaswani et al., 2017]. Following the cross-attention stage, multi-layer perceptron (MLP) layers apply non-linear transformation to the contextualised feature embeddings produced from the A2V and V2A mechanisms to refine the modality-enhanced features and learn higher-level feature interactions within each modality. Each MLP expands the embedding dimension from 768 to 3072 using an expansion ratio of 4, applies a Gaussian Error Linear Unit (GELU) activation function, and then projects the representation back to 768 dimensions.

Fusion Cross-Attention The A2V and V2A embeddings are concatenated along the feature dimension to form a joint multimodal representation [Cappellazzo et al., 2025]. This representation is processed using a secondary cross-attention module sharing the same architectural configuration as the bidirectional fusion modules. A learnable fusion token is introduced to aggregate contextual information from both modalities into a unified representation. This token acts as the query representation, while the concatenated embeddings provide the corresponding key and value representations. Through attention, the fusion token attends to the audio and visual information to create the final fused embedding.

To train the fusion module for multilabel object presence prediction, binary cross-entropy with logits was employed. The loss function enables independent probability estimation for each object category within the

multilabel prediction task. Binary cross-entropy with logits is defined as:

$$\mathcal{L}_{\text{BCE}}(\ell, y) = -[y \cdot \log(\sigma(\ell)) + (1 - y) \cdot \log(1 - \sigma(\ell))] \quad (3)$$

where ℓ represents the predicted logits, y denotes the ground truth label and $\sigma(\cdot)$ is the sigmoid activation function. While the SpokenCOCO subset is balanced for this paper, the original dataset is not balanced; a weighted version of the formula was used:

$$\mathcal{L}_{\text{BCE}} = -[w^+ \cdot y \cdot \log(\sigma(\ell)) + (1 - y) \cdot \log(1 - \sigma(\ell))] \quad (4)$$

where w^+ represents the positive class weighting factor. This weighting formulation is particularly relevant for scaling the proposed pipeline to the full SpokenCOCO dataset for future work, where greater class imbalance is present.

Grounding supervision was incorporated to encourage alignment between spoken object references and visually relevant regions. Grounding loss was applied to the A2V attention maps to encourage alignment between attended visual regions and the corresponding object annotations. The grounding formula is represented as:

$$\mathcal{L}_{\text{ground}} = f_{\text{ground}}(\text{Attn}_{a \rightarrow v}, \mathcal{B}) \quad (5)$$

where $\text{Attn}_{a \rightarrow v}$ denotes the audio-to-visual attention maps and $\mathcal{B}$ represents the corresponding grounding annotations.

The final training objective equation combines both losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ground}} + \lambda_{\text{joint}} \cdot \mathcal{L}_{\text{BCE}} \quad (6)$$

where λ_{joint} controls the relative contribution of the classification objective. This weighting factor prevents either loss component from dominating optimisation during training and enables balanced learning between localisation and semantic prediction tasks. Training was performed using the AdamW optimiser with a learning rate of 3×10^{-5} , weight decay of 0.1, batch size of 8, and 15 epochs. All experiments were conducted using an NVIDIA GeForce GTX 1080 GPU. The output representation of the fusion token was used as the final multimodal embedding for downstream classification.

Classifiers The fused multimodal embeddings were evaluated using two downstream classifiers: Support Vector Machines (SVMs) [Hearst et al., 1998] and Logistic Regression [Sperandei, 2014]. These classifiers were selected due to their effectiveness on high-dimensional embedding representations and their common use for evaluating pre-trained feature spaces without requiring end-to-end fine-tuning. Logistic Regression provides probabilistic linear classification, while SVMs maximise the separating margin between classes within the learned embedding space.

4 Results

To evaluate the proposed pipeline, the audio and visual modalities were first assessed independently to establish baseline performance. The fusion module was then evaluated in two configurations: an untrained configuration and a trained configuration using the hyperparameters selected in the Fusion Cross-Attention section. This comparison enabled evaluation of the impact of training the multimodal fusion module on overall performance.

Table 2 presents the classification performance of the proposed pipeline across single-modality and multimodal fusion configurations. The experiments compare audio-only embeddings, visual-only embeddings, multimodal untrained fusion, and multimodal trained fusion using the optimal hyperparameters obtained through grid search. Performance was evaluated using Micro F1, Macro F1, and mean Average Precision (mAP). Micro F1 measures overall classification performance using the global counts of true positives, false positives, and false negatives, while Macro F1 computes the unweighted average of per-class F1 scores, providing greater sensitivity to class-level performance differences. mAP evaluates ranking quality across classes by combining

precision and recall characteristics. A 5-fold Group KFold cross-validation strategy was used to reduce data leakage between training and validation splits by ensuring that samples associated with the same image were not simultaneously present in both partitions [Raschka, 2018].

The results demonstrate that multimodal fusion substantially outperforms single-modality representations across all evaluation metrics. Visual-only embeddings consistently achieved stronger performance than audio-only embeddings, indicating that the visual modality contains more discriminative information for the selected object categories. However, incorporating audio representations through cross-attention fusion further improved classification performance, demonstrating that speech-derived contextual cues provide complementary information beyond visual features alone.

Configuration	Classifiers	Micro F1	Macro F1	mAP
Audio-only	SVM	$73.912 \pm 0.315\%$	$70.180 \pm 0.367\%$	$77.524 \pm 0.631\%$
	Logistic Regression	$75.087 \pm 0.421\%$	$72.021 \pm 0.494\%$	$79.345 \pm 0.576\%$
Image-only	SVM	$84.496 \pm 0.103\%$	$82.592 \pm 0.249\%$	$88.744 \pm 0.215\%$
	Logistic Regression	$85.151 \pm 0.259\%$	$83.626 \pm 0.419\%$	$89.665 \pm 0.263\%$
Untrained Fusion	SVM	$89.574 \pm 0.386\%$	$88.709 \pm 0.567\%$	$92.724 \pm 0.809\%$
	Logistic Regression	$90.460 \pm 0.380\%$	$89.897 \pm 0.486\%$	$93.994 \pm 0.725\%$
Trained Fusion	SVM	$92.110 \pm 0.456\%$	$91.279 \pm 0.620\%$	$93.847 \pm 0.735\%$
	Logistic Regression	$92.160 \pm 0.500\%$	$91.380 \pm 0.650\%$	$94.131 \pm 0.660\%$

Table 2: Classification performance of single-modality and multimodal fusion configurations evaluated using Micro F1, Macro F1, and mAP.

Notably, even the untrained fusion configuration outperformed both individual modalities, suggesting that cross-modal contextual integration provides complementary information prior to optimisation of the fusion module. The relatively strong performance of the untrained fusion configuration suggests that the pre-trained transformer encoders already produce highly transferable modality-specific representations. Optimising the fusion module further improves cross-modal alignment and contextual integration, leading to the highest overall classification performance. Logistic regression achieved marginally higher performance than SVM across most configurations, suggesting that the fused embedding space produced by the transformer-based encoders was sufficiently linearly separable for lightweight probabilistic classification.

5 Conclusion and Future Work

The experimental results demonstrate that the proposed cross-attention fusion pipeline improves classification performance compared to both unimodal representations and multimodal untrained fusion configurations. The controlled, balanced subset enabled clearer evaluation of modality fusion behaviour while reducing the influence of severe class imbalance present in the original SpokenCOCO dataset. Experimental evaluation demonstrated that multimodal fusion consistently outperformed single-modality configurations across all evaluation metrics. The results further showed that integrating speech-derived contextual information with visual representations improved classification performance beyond visual-only embeddings, highlighting the complementary nature of audio-visual feature fusion. Additionally, the strong performance of the untrained fusion configuration suggests that pre-trained transformer encoders provide highly transferable modality-specific representations prior to fusion optimisation. Training the fusion module further improved cross-modal contextual alignment and overall classification performance. However, the use of a balanced subset limits direct comparison with state-of-the-art (SOTA) methods evaluated on the full SpokenCOCO dataset, where substantial class imbalance can significantly affect performance metrics. Future work will evaluate both the proposed pipeline and comparable SOTA architectures under full-dataset conditions to enable fairer comparative analysis. Further development will additionally explore end-to-end fine-tuning of the modality encoders, larger multimodal datasets, and more advanced fusion strategies for improving cross-modal grounding and contextual alignment.

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